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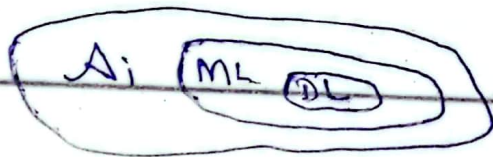
Deep Learning

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What is deep learning?

Deep learning is a subfield of AI and ML that is inspired by the structure of a human brain.

→ D.L algorithms attempt to draw similar conclusions as humans would by continually analyzing data with a given logical structure called neural network.



ML algorithms mostly focuses on

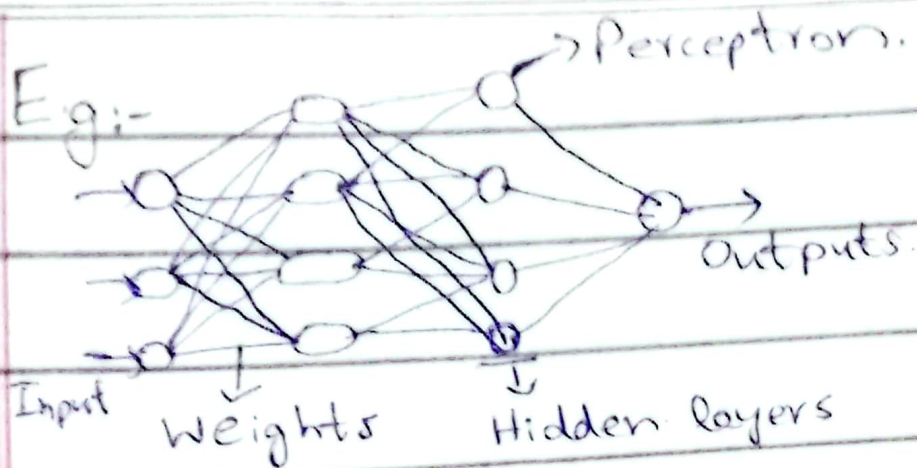
- statistical techniques, Whereas

D.L depends on a logical structure called neural network.

Basically a logical structure. ↳ Inspired by human brain.

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Name d.L is connected with these hidden layers that's why name D.L.

- ANN
- RNN (Recurrent neural network) → speech/text.
- CNN (convolutional " " → Image
- GAN → generate things.

Why d.L so famous:-

- 1) Applicability → Applied to wide domain of problem
- 2) Performance

→ DL is a part of ML methods based on a.NN with representation learning

↳
P.T.D.

Less data $\rightarrow$ ML wins.

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Representation Learning:-

Automatic feature extraction

- Feature engineering isn't needed in DL
e.g:- In ML Dog/cat classifier.
we featured on size, colour etc.

DL VS ML:-

- 1) Data dependency:- Data $\uparrow$ hungry
- 2) Hardware dependency:- $\rightarrow$ costly in DL.
 $\searrow$ Matrix multiplication $\rightarrow$ GPU
 $\downarrow$ Powerful

- 3) Training time:-
T.T is very high in D.L.

- 4) Feature selection:-

- 5) Inter-pretability:- Quite low & its flaw. (Black box)

In DL models its $\downarrow$ means
we can't provide solid proof that
why u ain't selected or why u banned.

step function.



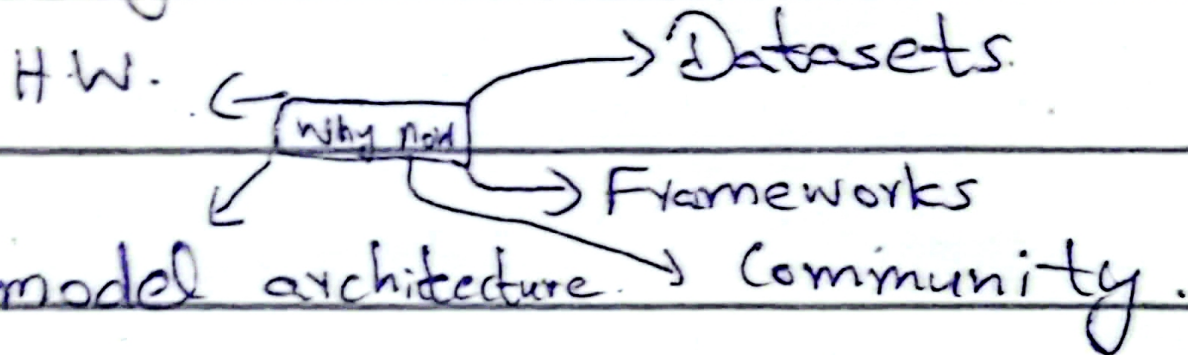
$$\begin{array}{r} 270 \ 1 \\ 240 \ 0 \end{array}$$

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So that's why DL can't replace ML.

Why now?



frame-
works.

Keras + Tensorflow $\rightarrow$ Industry driven application

Pytorch $\rightarrow$ Research purpose.